

Aiden Fox

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Education

University of Southern California (USC) — Los Angeles, CA

Aug 2024 – Dec 2027

BS Computer Engineering & Computer Science, GPA 3.60; E-Board Theta Tau, E-Board Beta Theta Pi, Treasurer QEE

Relevant coursework: Embedded Systems, Digital Circuits, Linear Circuits, Distributed Systems, Data Structures & Object-Oriented Design, Algorithms.

Stanford University — Summer Session, CompE & CS, GPA 3.66

2023

Experience

Software Engineering Intern, ML Team, Motif Inc. — Los Angeles, CA

May 2025 – Present

- Developed a deep-research agent in an AI product from 0 to 1, then architected it into a reusable, model-agnostic framework other skills compose against, and made it durable and async-resumable via a workflow-backed runtime.
- Scaled an OCR-routed SageMaker inference pipeline across 22,000+ images via multi-GPU EC2 sharding, cutting batch-processing time ~65%.
- Refactored a production backend to AWS Lambda and SQS with retries and CloudWatch observability.

Projects

Fennec Autonomous Quadruped

Teensy 4.1, micro-ROS, ROS 2 Humble, LIDAR-Inertial SLAM

(built on the open-source Nova-SM3 platform, heavily redesigned)

- Engineered a 200 Hz IntervalTimer ISR-driven real-time loop on Teensy 4.1 (IMXRT1062 Cortex-M7) achieving p99 = 1 microsecond scheduling jitter, 100x under the <100 microsecond gate; added a software watchdog (SCB_AIRCR reset on hang) and latency histograms over USB-CDC.
- Built a Feitech STS3215 half-duplex bus driver from scratch (PING / READ_DATA / SYNC_WRITE, 74HC125 OE timing verified on a logic analyzer) with categorized error counters, a per-joint slew-rate limiter, and round-robin per-joint telemetry.
- Implemented a 22-topic ROS 2 contract over micro-ROS / XRCE-DDS to an NVIDIA Jetson Orin Nano (embedded Linux), with a latching E-stop and battery-low safety FSM (50 ms debounce, /safety_clear reset).
- Brought up LIDAR-Inertial SLAM (Unitree L2 and POINT-LIO) on Jetson, diagnosed and patched an IMU bridge bug in unilidar_sdk2 (double getImuData() drain), and released a public fork that restored the 250 Hz IMU stream.

Fennec Power Distribution Board

KiCad, Multi-Rail, INA226 Telemetry

- Designed a custom multi-rail power PCB (battery to buck rails) with INA226 per-rail current/voltage telemetry, MOSFET reverse-polarity protection, and a hardware safety/E-stop chain, then ran DRC/ERC clean and fabricated via JLCPCB.
- Sized rails against a measured power budget (servo stall transients; Jetson and LiDAR), selected per-rail INA226 shunts feeding the firmware's battery-low shutdown FSM, and put the servo-bus 74HC125 driver stage on-board.

6-DOF Robotic Arm

Raspberry Pi, TTL Serial Bus, SmolVLA, MuJoCo

- Drove 6x Feitech STS3215 over a 1 Mbaud TTL bus, fine-tuned a SmolVLA visuomotor policy (105 demos, 11 tasks) with local STT for voice-commanded manipulation, and validated closed-form IK in MuJoCo (0.0 mm FK round-trip error).

GitSpider (gitspider.com)

TypeScript, Docker

- Shipped a live GitHub App that flags flaky runs, slowdowns, and wasted compute in GitHub Actions workflows via structural YAML analysis.

Technical Skills

Embedded & Real-Time: Teensy 4.x (IMXRT1062 Cortex-M7), bare-metal C/C++, RTOS, IntervalTimer ISR control, 200 Hz loops with sub-microsecond jitter, software watchdog (SCB_AIRCR), half-duplex serial bus drivers, I2C, SPI, UART, TTL, USB-CDC, Ethernet (DDS), ADC, PID control, oscilloscopes, logic analyzers

Robotics & Autonomy: ROS 2 (Humble), micro-ROS, XRCE-DDS, LIDAR-Inertial SLAM (POINT-LIO), NVIDIA Jetson Orin Nano, sensor fusion, safety FSMs / latching E-stop, Feitech STS3215, PlatformIO

Hardware: KiCad, multi-rail power PCB design, INA226 rail telemetry, MOSFET reverse-protection, buck converters, FPGA (Xilinx), digital logic (FSMs, datapaths, control units)

Languages: C++, C, Python, Verilog, Java, JavaScript, TypeScript

ML & Systems: PyTorch, SmolVLA (VLA policies), CUDA, AWS (EC2, SageMaker, Lambda), Docker, FastAPI, React

Tools: Linux CLI, Git, CI/CD (GitHub Actions)